

Preliminary: exploratory calculation, not a validated result

These calculations are two-dimensional, and this is the measurement rather than the claim. The mesh is one cell deep, and the two faces that close the span carry no solution values of their own, so the flow has no room to vary along the span. Every cell was matched face to face across the span by position and every solved quantity compared: the differences are zero. The right-hand chart asks the same question physically — how fast does air move along the span? Below $1\times10^{-15}\text{ m s}^{-1}$, the level at which the arithmetic itself stops being able to tell a number from zero.

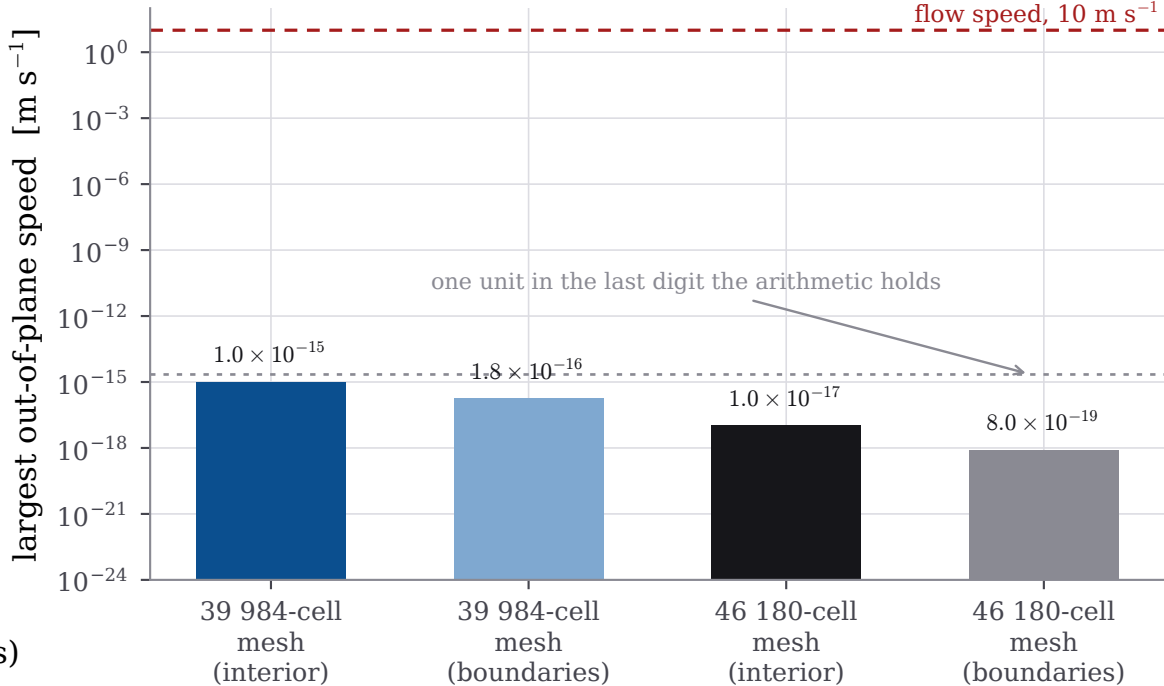
Every solved quantity, measured end to end across the span



largest difference between the two ends of the span (in each quantity's own units)

● 39 984-cell mesh — measured (worst of five) ■ 46 180-cell mesh — measured ○ the same check on a deliberately altered copy

Air motion along the span: how much is there?



MEASURED VALUES BEHIND THIS CHART (39 984 and 46 180 cells; one face pair compared per cell, 39 984 and 46 180 pairs)					
quantity	units	largest span difference, 39 984-cell mesh	largest span difference, 46 180-cell mesh	as a fraction of the largest value in the field [-]	the same check on a deliberately altered copy
velocity	m s^{-1}	0 exactly	0 exactly	0 exactly	9.97×10^0
pressure per unit density	$\text{m}^2 \text{s}^{-2}$	0 exactly	0 exactly	0 exactly	6.66×10^0
turbulent kinetic energy	$\text{m}^2 \text{s}^{-2}$	0 exactly	0 exactly	0 exactly	3.24×10^{-1}
turbulence frequency	s^{-1}	0 exactly	0 exactly	0 exactly	1.27×10^8
turbulent viscosity	$\text{m}^2 \text{s}^{-1}$	0 exactly	0 exactly	0 exactly	4.65×10^{-4}
out-of-plane velocity component	m s^{-1}	1.02×10^{-15}	1.02×10^{-17}	1.02×10^{-16}	— (an absolute level, not a difference)

WHAT YOU SHOULD KNOW ABOUT THESE NUMBERS

- A zero is only worth something if the check could have found something else. Every comparison was repeated on a copy in which the pairing between the two ends of the span was shifted by one cell on purpose. On those copies the check reports large differences — the red circles, and the last table column. The zeros are therefore a finding, not a blind spot.
- Two meshes are reported side by side and never mixed: 39 984 cells with a 0.01 m span, and 46 180 cells with a 1 m span. No lift or drag figure is formed from either on this chart.
- The differences are exactly zero, not merely small — which is what a one-cell-deep calculation should give, and the point of the check: the calculation that ran is the two-dimensional one that was intended.
- Air speed ALONG the span is not identically zero and we do not claim it is. It reaches $1.0\times10^{-15}\text{ m s}^{-1}$ in the coarser mesh and $1.0\times10^{-17}\text{ m s}^{-1}$ in the finer one — 1×10^{-16} and 1×10^{-18} of the flow speed, at or below one unit in the last digit the arithmetic holds. Rounding, not motion.
- This check spent no computer time of its own; it re-reads results already produced. The convergence caveats on the other charts apply unchanged: none of these calculations met the target that was fixed before they ran.